10/01 Prostate MRI Exploratory Data Analysis (EDA)

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Dataset Overview

Source: https://liuquande.github.io/SAML/
Description: prostate T2-weighted MRI data

Dataset	Institution	Case num	Field strength(T)	Resolution(in/ through plane)(mm)	Endorectal Coil	Manufactor
Site A	RUNMC	30	3	0.6-0.625/3.6-4	Surface	Siemens
Site B	BMC	30	1.5	0.4/3	Endorectal	Philips
Site C	HCRUDB	19	3	0.67-0.79/1.25	No	Siemens
Site D	UCL	13	1.5 and 3	0.325-0.625/3-3.6	No	Siemens
Site E	BIDMC	12	3	0.25/2.2-3	Endorectal	GE
Site F	HK	12	1.5	0.625/3.6	Endorectal	Siemens

source: (Liu, Quande, et al.) Table 1

Dataset Inventory

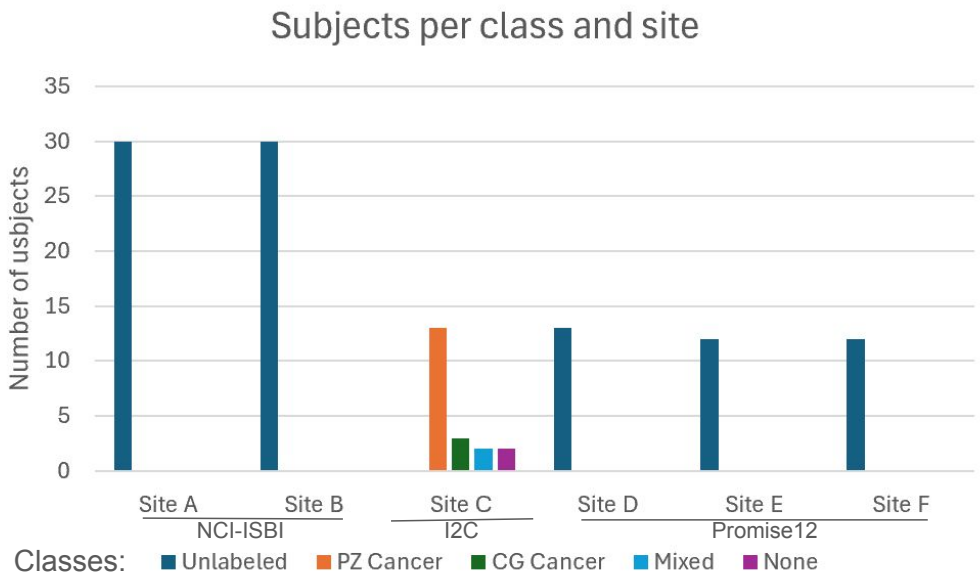


Figure 1.

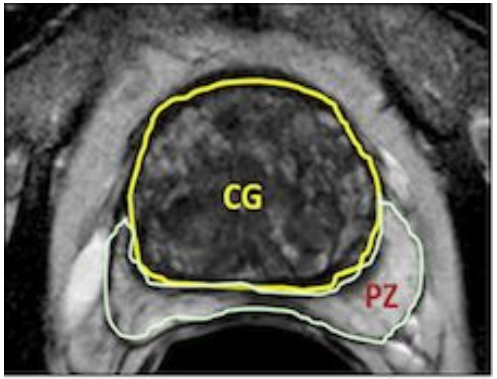
Although all sites contain annotated images, only Site C includes patient disease status.

Sites A and B each contain more than twice as many patients as Sites D-F.

Site C contains only 2 non-disease cases, resulting in class imbalance.

Image Visualization and Format (pixel labels)

source: (NCI-ISBI 2013 Challenge)



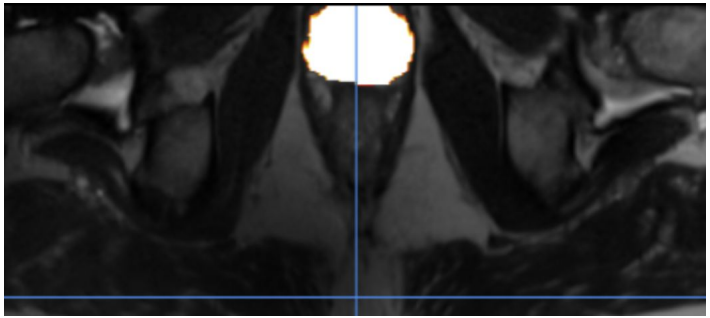
Sites A, B (NCI-ISBI):

CG, PZ, and background

Labels: 0 = background, 1 = CG, 2 = PZ

Provides zonal prostate segmentation

visualization: (used neuropsis.org/nifti_viewer.html)



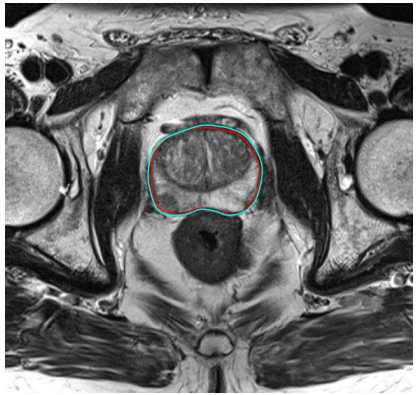
Site C (I2CVB):

Original annotations include CG, PZ, prostate gland, and CaP lesions

Processed masks used here: 0 = background, 1 = prostate foreground

Therefore treated as binary whole-prostate segmentation in the model

source: (MICCAI Grand…)



Sites D,E,F (Promise12):

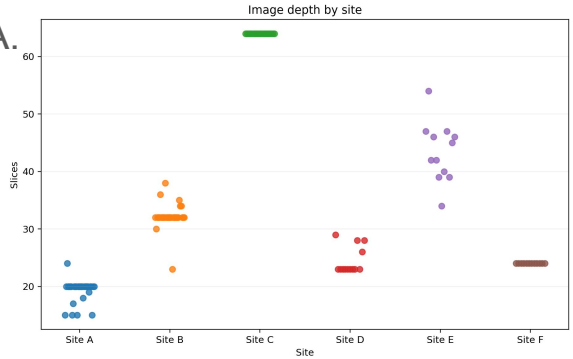
Whole prostate and background

Labels: 0 = background, 1 = prostate

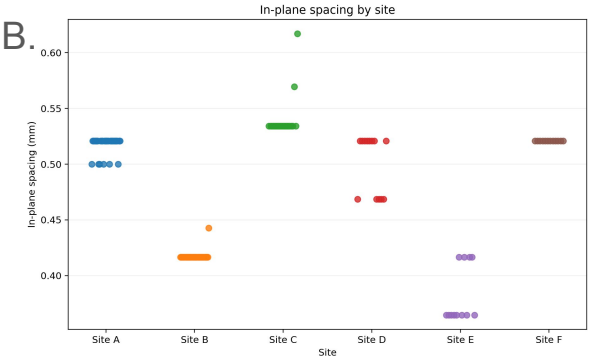
Binary segmentation → simplest annotation scheme

Image geometry per site

A.



B.



C.

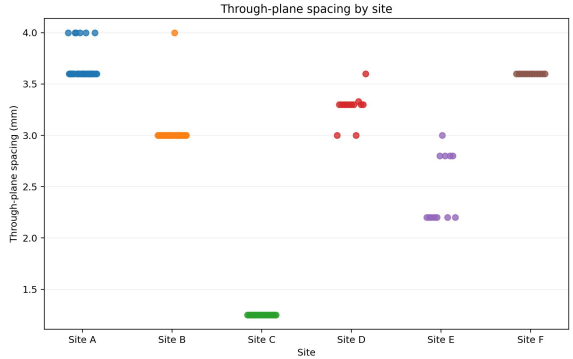


Figure 2.

Image dimensions across all sites are 384 x 384. 2D dimensions are consistent.

Image depth varies, with Site C having substantially more slices, so padding, cropping, or resample is needed (1A).

Voxel spacing varies across sites, especially D-E, with several outliers (1B-C). Resampling will be needed.

These geometry differences test the generalization across sites.

Intensity (brightness) distribution per site

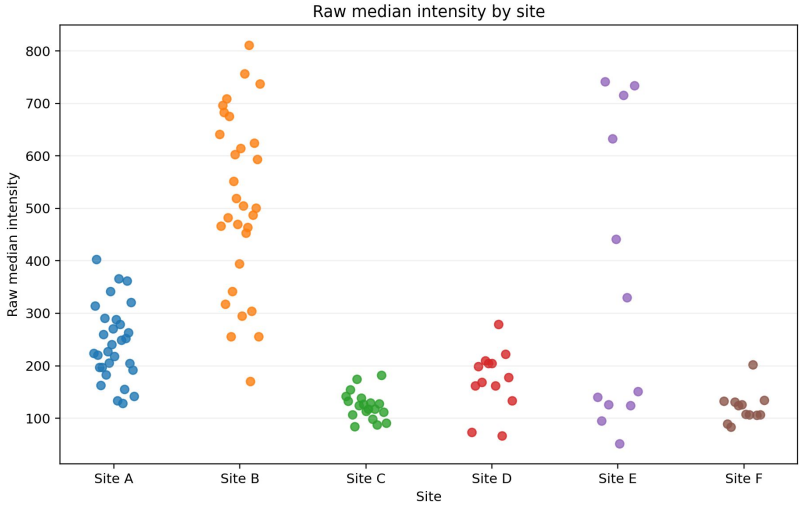


Figure 3.

Field strength(T)
3
1.5
3
1.5 and 3
3
1.5

Endorectal Coil	Manufactor
Surface	Siemens
Endorectal	Philips
No	Siemens
No	Siemens
Endorectal	GE
Endorectal	Siemens

Large inter-image/site intensity variation → normalization needed.

Higher field strength generally produces more MRI signal, improving SNR though raw MRI intensity is not standardized.

Coil configuration may contribute to variation, but does not fully explain site-level differences.

Scanner manufacturer may also contribute: Siemens scans generally show lower intensities, while Philips/GE scans show higher and more variable intensities.

Segmentation and pixel labels per site

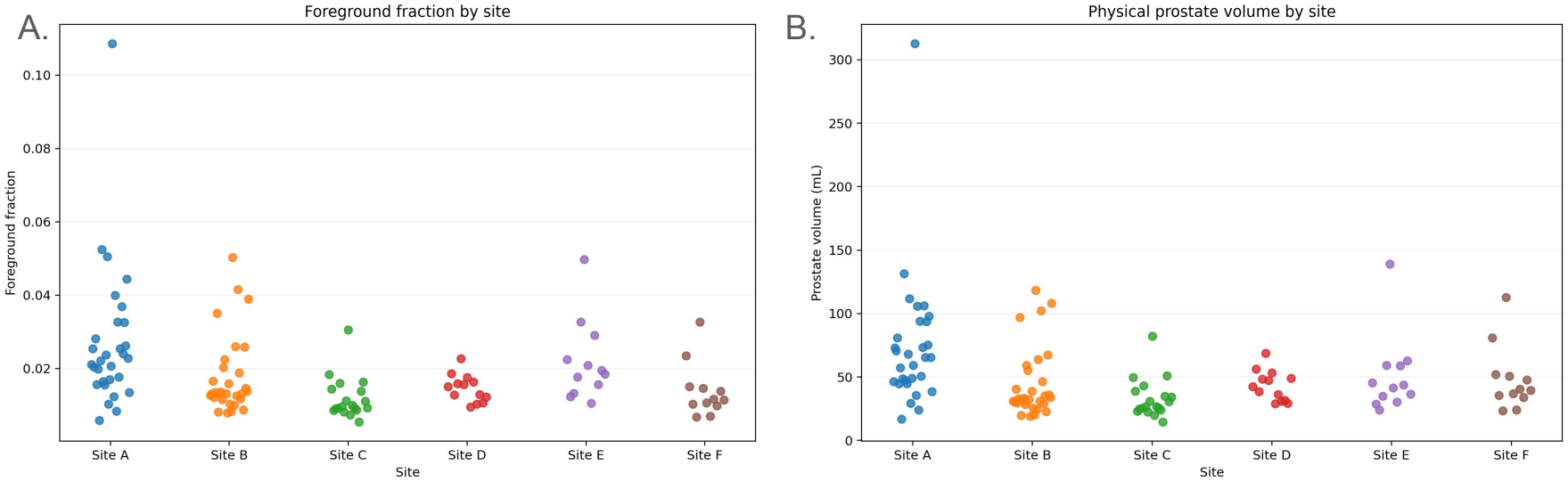


Figure 4.

1. Prostate foreground fraction remains relatively consistent at ~2%, supporting limited need for cropping. Spatial alignment requires further investigation.
2. Prostate foreground fraction and physical prostate volume show similar within- and across-site patterns, supporting the model's biological consistency.

Questions?

References

Liu, Quande, et al. “FedDG: Federated Domain Generalization on Medical Image Segmentation via Episodic Learning in Continuous Frequency Space.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 1013–1023.

https://wiki.cancerimagingarchive.net/display/Public/NCI-ISBI+2013+Challenge+-+Automated+Segmentation+of+Prostate+Structures

https://i2cvb.github.io/

https://promise12.grand-challenge.org/